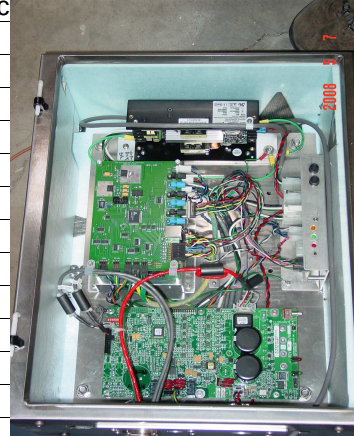


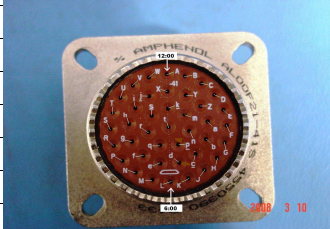
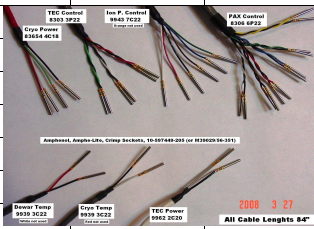
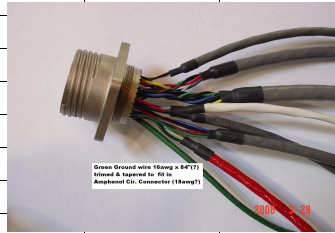
30-27-503-B Rim Box Wiring Schedule							
File name:	30-27-502-B Rim_Wiring.xls						
Created:	2005-02-29 Dave Deboer						
Updated	2007-02-22 Gary Gimblin						
Updated	2007-05-02 Matt Fleming (Rev B new 41 pin connector)						
					datasheet-PAX.doc		
Related Documents							
	30-27-502 Rim Box Wiring Diagram. (primary reference document)						
	30-27-503 Rim Box Wiring Schedule. (this doument)(excel file)						
	30-27-520 Rim Power & Ion Monitor Board.						
	30-27-530 Rim Control Board.						
	30-27-540 Rim Power Management Board.						
	30-27-550 TEC Control Board.						
	30-27-560 STI Cryo Control Board.						
	30-27-600 Front End Wiring Diagram.						
	30-24-360 Ion Pump Control (mechanicals)						
	30-24-370 Ion Pump Monitor (mechanicals)						
	30-26-230 Cable Assy, Rim to Pax.						
	30-26-234 Connector Amphenol AL00F21-41S (IP)						
Notes:							
	Data in tables supperceeds info in photos.						
	AR1 is listed here just for reference. (actual specs are part of alidade)						
	RW1 thru RW13 are listed below						
	RF1 is also listed here for convenience.						
	RP1 no longer exists. It was replaced as RW1 was extended with a new connector.						
	For hookup wire try to get 22 awg 19 x 34 or 20 awg 19 x 32. Avoid 7 x N						
	Cables RW1E, RW1F, RW1G, RW1J require various ferrites. (see drawing)						
	The letters I and O are not used as identifiers						
	Many related Mouser catalog pages & connector specs are saved in section 27 directories.						
AR1 from Alidade 120 VAC to Rim Box Power Distribution Board.							
A	Description		Size	Color	Twist	Length	Comment
	Line 120VAC	B1	14 ?	P-B1	Z		
	Neutral	B2		P-W1	Z		
	Ground	B3					
B	50-84-1030	Molex MLX 0.084 dia, Plug Housing, 3pos, 1x3.					
	02-08-1004	Molex, MLX 0.084 dia, sockets 02-08-1004, 14-20 awg, bronze Sn.					
misc	Fuses are also needed.						

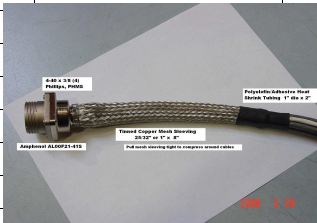
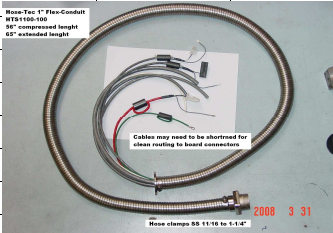
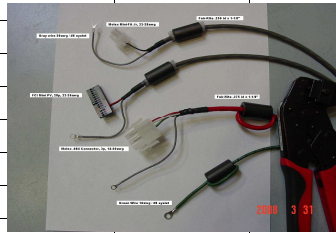


RW1 from PAX extension to various boards and components inside Rim Box.							
A	Description	B,C,D,E,F,G	Size	Color	Twist	Length	Comment
Z	GND (OTXTEC)	B1	6P22 Bel 8306 Foil & Braid	black	Z	84"	RW1B B13 shield ?
Y	+5 (OTXTEC)	B2		red	Z		
B	GND (OTXlaser/PAM)	B3		black	Y		
C	+6 (OTXlaser/PAM)	B4		yellow	Y		
k	GNDsense	B5		black			
s	+6sense	B6		green			
X	+5sense	B7		white			
m	-6sense	B8		black			
j	+5 (OTXTEC)	B9		brown	X		
i	GND (OTXTEC)	B10		black	X		
W	-6 (OTX/PAM)	B11		blue	W		
V	GND (OTXPAM)	B12		black	W		
	PAX Control shield	term eye	20 ?	gray		short	
	TEC Control shield	term eye	20 ?	gray		short	C1 shield ?
b	PAX thermistor 1	C2	3P22 Bel 8303	black	U	84"	RW1C
H	PAX thermistor 2	C9		white	U		
c	Airstream thermistor	C3		black	S		
J	Airstream thermistor	C10		green	S		
a	Heat sink thermistor	C4		black	T		
G	Heat sink thermistor	C11		red	T		
q	TEC power -	C8	2C20	black		84"	RW1D
P	TEC power +	C15	Bel 9962	gray			
(RW5)	Power 24 com	C6	20	black	R	short	RW5
(RW5)	Power 24 +	C13	20	white	R		
U	Dewar temp 1	F21	3C22	red		76"	RW1F
T	Dewar temp 2	F22	Bel 9939	black		white cut	
	Dewar temp shield	term eye	20	gray		3"	
K	Cryocooler power +	E1	4C18 Bel 83654	red		76"	RW1E
d	Cryocooler power +			white			
M	Cryocooler power -	E2		black			
e	Cryocooler power -			green			
	Cryocooler shield	E3	20	gray		3"	
E	Cryo heatsink therm	G1	3C22	white		76"	RW1G
D	Cryo heatsink therm	G2	Bel 9939	black		red cut	
	Cryo heatsink shield	term eye	20	gray		short	
f	Current monitor	H1	7C22 Bel 9943 or 6C24 Bel 9611	blue		84" orange cut	RW1H
g	External reset	H3		white			
h	Power + (12 VDC)	H4		red			
r	Power -	H5		black			
P	Fault high voltage	H7		brown			
R	Fault blown fuse	H8		green			
	Ion control shield	H6	20	gray		short	
L	Ground case to case	term eye	16	green		76"	RW1J
A,N,n,S,t not used for shield carry through.							

See also drawing 30-27-502

RW1	from PAX extension to various boards and components inside Rim Box. (continued)		
A	AL00F21-41S (IP)	Amphenol, flange mount, 41 soc, EMI tight, kit with sockets. (30-26-234)	
B, C, D	205206-3	AMP, HDP-20 D-sub 15, pin, crimp snap -in, w indents, tin plt.	
	6-66506-1	AMP, HDP-20 Pin, crimp, 20-24, brass, gold, 100/pk.	
	1-66506-0 ?	AMP, HDP-20 Pin, crimp, 20-24, brass, gold, 100/pk.	
(RW5)	39-01-2020	Molex, Mini-Fit Jr, Recpt Housing, 2 pos, Dual Row.	
	39-00-0059	Molex, Mini-Fit Jr, Socket, 18-24awg.	
E	50-84-1030	Molex MLX 0.084 dia, Plug Housing, 3pos, 1x3.	
	02-08-1004	Molex, MLX 0.084 dia, sockets 02-08-1004, 14-20 awg, bronze Sn.	
F	69176-028LF	FCI, Mini-Latch, housing, 28 pos, 2 row, 0.100 sp, keyed, black.	
	48254-000LF	FCI, Mini-Pv, contact receptical, 22-26awg FCI	
G	39-01-2020	Molex, Mini-Fit Jr, Recpt Housing, 2 pos, Dual Row.	
	39-00-0041	Molex, Mini-Fit Jr, socket, 18-24awg, brass, Sn.	
	39-00-0060 (?)	Molex, Mini-Fit Jr, socket, 18-24awg, bronze, Sn, bag.	
	39-00-0182 (alt)	Molex, Mini-Fit Jr, socket, 18-24awg, bronze, Au.	
H	43025-0800	Molex, Micro-Fit, Recpt Housing, 8 pos, Dual Row.	
	43030-009	Molex, Micro-Fit, contact, soc, 20-24 awg, brass, gold 30μ, /bag	
J	????	Terminal, eye, 16 awg #8, crimp on. (main case to case ground)	

		
RW1A Photo-01	RW1A photo-02	RW1A Photo-03

		
RW1A Photo-04	RW1A Photo-05	RW1A Photo-06

RW2 from Power Supply 24VDC to STI Cryo Control Board							
A	Description	B	Size	Color	Twist	Length	Comment
1	24 VDC pos	1	16 ?	red			
2	Common	2		black			
A, B	50-84-1020	Molex, MLX 0.084 dia, Plug Housing, 2pos, 2 row, 1 col.					
	02-08-1004	Molex, MLX 0.084 dia, sockets, 14-20 awg, bronze, Sn.					
A, B (alt)	1-480698 ?	Tyco, Plug housing, 2x1, (newark 46F2063) (industry standard Mate-N-Loc					
	350550-2 ?	Tyco, contacts, (Newark 90F5656)					
RW3 from STI RS232 to Rim Control							
A	Description	B	Size	Color	Twist	Length	Comment
10	Ground	4 LL	22 ?	green	or 1	?	
11 LR	RxD TxD	5		yellow	or 2		
12 UR	TxD RxD	6 LR		blue	or 3		
A	69176-012LF	FCI, Mini-Latch, housing, 12 pos, 2 row, 0.100 sp, keyed, black.					
	48254-000LF	FCI, Mini-Pv, contact receptical, 22-26awg.					
B	22-55-2061	Molex, SL, housing, 6 pos, 2 row, non-polarized.					
	16-02-0102	Molex, SL, contact, socket. 22-24 awg, bronze, Sn150, 3.0 amp.					
	16-02-1109 (alt)	Molex, SL, contact, socket. 22-24 awg, bronze, Au15, high force, 3.0 amp.					
B	87499-5 (alt)	AMP, MOD.IV, housing, 3 pos, 1 row, 0.100 sp, non polerized.					
	2-87523-4 (alt)	AMP, MOD.IV, contact, receptacle, 24-20 awg, bronze, Au15. 1.5oz pull.					
	1-87309-3 (alt)	AMP, MOD.V, contact, receptacle, 24-20 awg, bronze, Sn100. 3.0oz pull.					
B	Note: Numbers on board are not pin numbers. They are for RS-232 unit 1 and unit 2						
RW4 from TEC Control to Rim Control							
A	Description	B & C	Size	Color	Twist	Length	Comment
1	(not used)		22 ?		colors may change	12" ?	
2	Rim box heatsink	B5 ADC-7		green			
6	Ground	B4		green			
3	PAX thermistor	B2 ADC-8		green			
7	Ground	B1		green			
4	Airstream therm	C5 ADC-9		yellow			
8	Ground	C4		yellow			
5	TEC heatsink	C2 ADC-10		yellow			
9	Ground	C1		yellow			
A	205204-4	AMP, HDP-20 D-sub 9, pin, crimp snap -in, w indents, tin plt.					
	6-66506-1	AMP, HDP-20 Pin, crimp, 20-24, brass, gold, 100/pk.					
B, C	22-55-2061	Molex, SL, housing, 6 pos, 2 row, non-polarized.					
	16-02-0102	Molex, SL, contact, socket. 22-24 awg, bronze, Sn150, 3.0 amp.					
	16-02-1109 (alt)	Molex, SL, contact, socket. 22-24 awg, bronze, Au15, high force, 3.0 amp.					
B, C	87456-2 (alt)	AMP, MOD.IV, housing, 6 pos, 2 row, 0.100 sp, non polerized.					
	2-87523-4 (alt)	AMP, MOD.IV, contact, receptacle, 24-20 awg, bronze, Au15. 1.5oz pull.					
	1-87309-3 (alt)	AMP, MOD.V, contact, receptacle, 24-20 awg, bronze, Sn100. 3.0oz pull.					
B, C	Consider redesign board for Molex mico-fit headers & socket. (keyed & latched)						

RW5 from 24V Power Distrib Board to TEC Board							
A	Description	B	Size	Color	Twist	Length	
	(listed as part of RW1 above)						
RW6 from Power Distribution 12 VDC to Power Manage & Rim Control (12 VDC from Power Manage							
A	Description	B & C	Size	Color	Twist	Length	
1 UR	12 VDC pos	B1	20 ?	red blue		?	
3 LR	Com	B2		black			
2 UL	12 VDC pos	C1		red blue			
4 LL	Com	C2		black			
A	39-01-2040	Molex Mini-Fit Jr, recpt housing, 4 pos, 2 row, 94V-2, no strain rel.					
	39-00-0041	Molex, Mini-Fit Jr, socket, 18-24awg, brass, Sn.					
	39-00-0060 (?)	Molex, Mini-Fit Jr, socket, 18-24awg, bronze, Sn, bag.					
	39-00-0182 (alt)	Molex, Mini-Fit Jr, socket, 18-24awg, bronze, Au.					
B, C	770849-2	Tyco AMP SL-156, 2 pos, housing.					
	770522-1	Tyco AMP SL-156, socket, size?, bronze?, Sn?.					
RW7 from Power Distribution 24 VDC to Power Manage Board (should use different AMP conn							
A	Description	B	Size	Color	Twist	Length	
1	24 VDC pos	1	20 ?	red		?	
2	Common	2		white			
A	39-01-2020	Molex Mini-Fit Jr, recpt housing, 2 pos, 2 row, 94V-2, no strain rel.					
	39-00-0041	Molex, Mini-Fit Jr, socket, 18-24awg, brass, Sn.					
	39-00-0060 (?)	Molex, Mini-Fit Jr, socket, 18-24awg, bronze, Sn, bag.					
	39-00-0182 (alt)	Molex, Mini-Fit Jr, socket, 18-24awg, bronze, Au.					
B	770849-2	Tyco AMP SL-156, 2 pos, housing.					
	770522-1	Tyco AMP SL-156, socket, size?, bronze?, Sn?.					
	770522-2 ?	Tyco AMP SL-156, socket, size?, bronze?, Sn?.					
	3-770522-1 ?	Tyco AMP SL-156, socket, size?, bronze?, Sn?, loose.					
RW8 from Power Manage to Focus connector (see RF1 Rim to Focus bottom of page)							
A	Description	B & C	Size	Color	Twist	Length	
1	Motor fwd ?	B2	22 ?	red		?	
2	Nutral	B3		white			
3	Motor fwd ?	B1 R		yellow			
4	Extend stop	C2 ?		orange			
5	Common	C1 R		brown			
6	Retract stop	C3 L ?		blue			
7	Ground	B4 L ?		green			
A	T 3478 009	Amphenol T 3478 009, panel mnt, 7 pin din, shielded.					
B	39860-0104	Molex, Beau, 4 pos, 0.200 sp. (115 VAC)					
C	43645-0300	Molex, Micro-Fit, Recpt Housing, 3 pos, single row.					
	43030-0009	Molex, Micro-Fit, contact, soc, 20-24 awg, bronze, gold 30μ, /bag					
	43030-0008	Molex, Micro-Fit, contact, soc, 20-24 awg, bronze, gold 15μ, /bag					

RW9 from Rim Control to Power Manage Board							
A	Description	B	Size	Color	Twist	Length	
19	Ground	1	22 ?	black		9" ??	
4	Out	2		green			
7	60 Hz	3		yellow			
17	In	4		white			
11	5 VDC	5		red			
13 & 14	Jumpered			Blue			
A	22-55-2201	Molex, SL, housing, 20 pos, 2 row, non-polarized.					
B	22-55-9005	Molex, SL, housing, 5 pos, 1 row, non-polarized.					
A B	16-02-0102	Molex, SL, contact, socket. 22-24 awg, bronze, Sn150, 3.0 amp.					
	16-02-1109 (alt)	Molex, SL, contact, socket. 22-24 awg, bronze, Au15, high force, 3.0 amp.					
RW10 from Power Distribution 120VAC to Power Manage Board							
A	Description	B	Size	Color	Twist	Length	
1	Line	1 R	16 ?	black		?	
2	Neutral	2		white			
3	Ground	3 L		green			
A	50-84-1030	Molex, MLX 0.084 dia, Plug Housing, 3pos, 1x3.					
	02-08-1004	Molex, MLX 0.084 dia, sockets 02-08-1004, 14-20 awg, bronze Sn.					
B	39860-0103	Molex, Euro, 3 pos					
RW11 from Power Distribution 120VAC to Auxillary (maybe not used)							
A	Description	B	Size	Color	Twist	Length	
1	Line 120 VAC		16 ?	black		?	
2	Neutral			white			
3	Ground			green			
A	50-84-1030	Molex, MLX 0.084 dia, Plug Housing, 3pos, 1x3.					
	02-08-1004	Molex, MLX 0.084 dia, sockets 02-08-1004, 14-20 awg, bronze Sn.					
B	Undetermined.	Undetermined.					
RW12 from Power Distribution 120VAC to Power Supply							
A	Description	B	Size	Color	Twist	Length	
1	Line	o	16 ?	black		?	
2	Neutral	o		white			
3	Ground	o		green			
A	50-84-1030	Molex, MLX 0.084 dia, Plug Housing, 3pos, 1x3.					
	02-08-1004	Molex, MLX 0.084 dia, sockets 02-08-1004, 14-20 awg, bronze Sn.					
B		Crimp terminal, fork, #8, size?awg.					

RW13 from Power Supply 24VDC to Power Distribution Board.							
A	Description	B	Size	Color	Twist	Length	
1	24 VDC pos	1	16 ?	red		?	
1	Com	2		black			
A	50-84-1020	Molex, MLX 0.084 dia, Plug Housing, 2pos, 1x2.					
	02-08-1004	Molex, MLX 0.084 dia, sockets 02-08-1004, 14-20 awg, bronze Sn.					
B	Undetermined.	Crimp terminal, fork, #8, size?awg.					
RW14 from Power & Ion Monitor to Rim Control.							
A	Description	B & C	Size	Color	Twist	Length	
1 LR	Current monitor	B2 ADC-12	22 ?	orange		?	
2 LC	Current mon gnd	B1		black			
5 UC	HV fault	B5 ADC-11		green			
4 UR	HV fault gnd	B4		black			
3 LL	External reset	C4		violet			
6 UL	Ext reset gnd	C8		black			
A	43025-0600	Molex, Micro-Fit, Recpt Housing, 6 pos, Dual Row.					
	43030-009	Molex, Micro-Fit, contact, soc, 20-24 awg, brass, gold 30μ, /bag					
B	22-55-2061	Molex, SL, housing, 6 pos, 2 row,					
C	22-55-2081	Molex, SL, housing, 8 pos, 2 row,					
B C	16-02-0102	Molex, SL, contact, socket. 22-24 awg, bronze, Sn150, 3.0 amp.					
	16-02-1109 (alt)	Molex, SL, contact, socket. 22-24 awg, bronze, Au15, high force, 3.0 amp.					
B	87456-2 (alt)	AMP, 6 pos, (similar to Molex above)					
C	87456-4 (alt)	AMP, 8 pos,					
RF1 Cable from rim to feed focus drive (see 30-25-130 Feed Focus Motor Assy)							
A	Description	B	Size	Color	Twist	Length	
1	Motor		7C22 Bel 9943	red		58"	
2	Motor			white			
3	Motor			black			
4	Extend stop			orange			
5	Common			brown			
6	Retract stop			blue			
7	Ground			green			
		Rim box end	13/16" strip back on connector 3/8" strip on shield				
		Terminal end	5" strip back on jacket/shield green #6 spade				